Input

1 Mb DNA
one-hot
hg38

Conv tower

11 RC conv
blocks, pool 2
→ 512 bins

Bi-Mamba ×8

RC-shared
scans
+ FiLM(cell)

Mixer

3 depths ×
4 scales
(1–8 bins)

Pair head

biaffine
+ b(c,|i−j|)
→ 448²

Objective

L_contact: masked MSE on log O/E
+ λ_ins L_ins + λ_bnd L_bnd
(last 25 % of steps)
insulation of g(C), g = exp(2 tanh(·/2)),
same operator on Ĉ and C

By construction

f(RC x, c) = rot180 f(x, c)
(unit test, fp32)
Ĉ symmetric
cell line enters via FiLM
+ distance bias only

Target (HFF, validation)
chr14:66.50–67.55 Mb



